

# IAS PRELIMS/MAINS



# SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

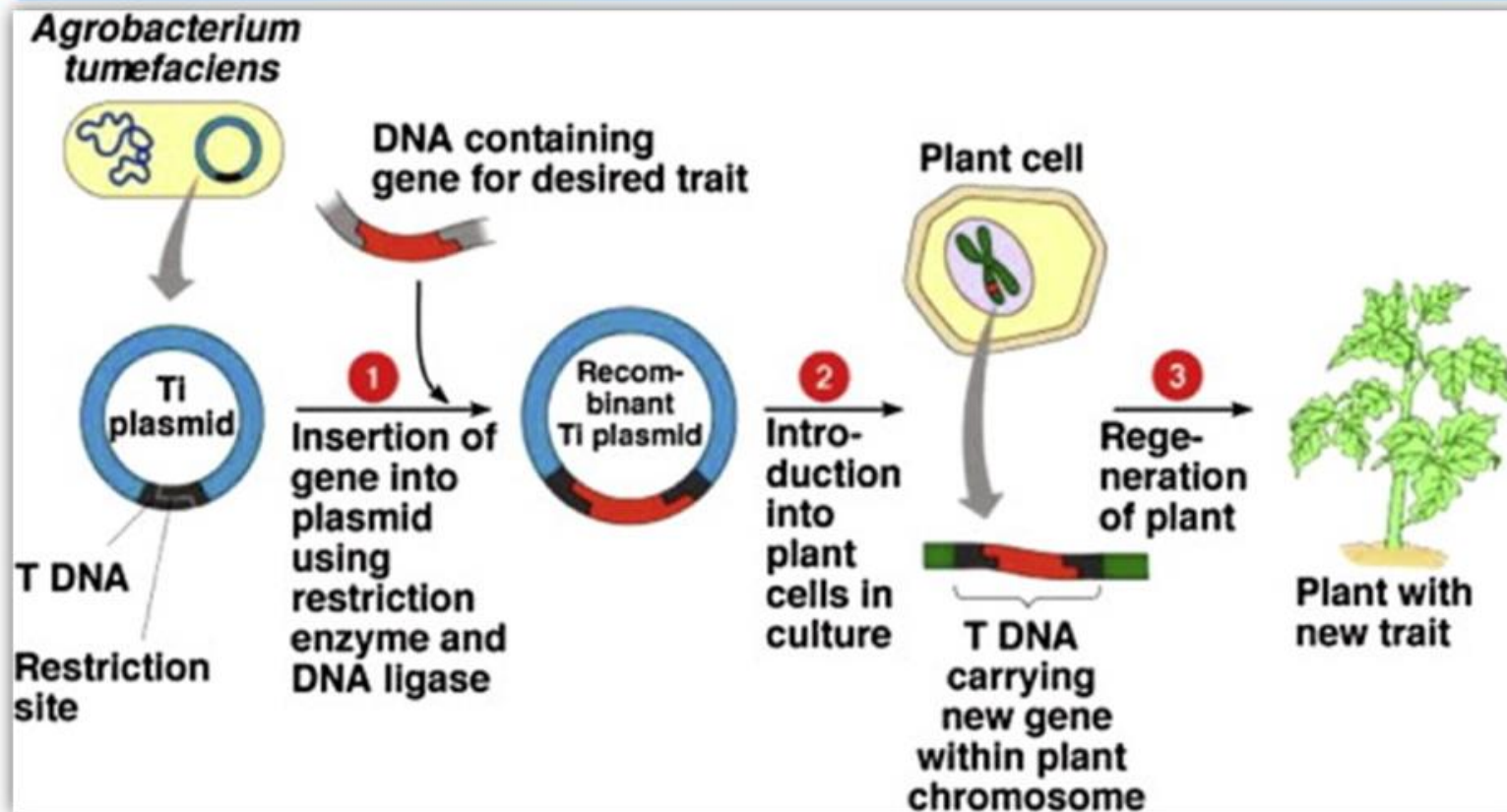
MODULE - 16

BIOTECHNOLOGY - 6



## TRANSGENIC PLANTS

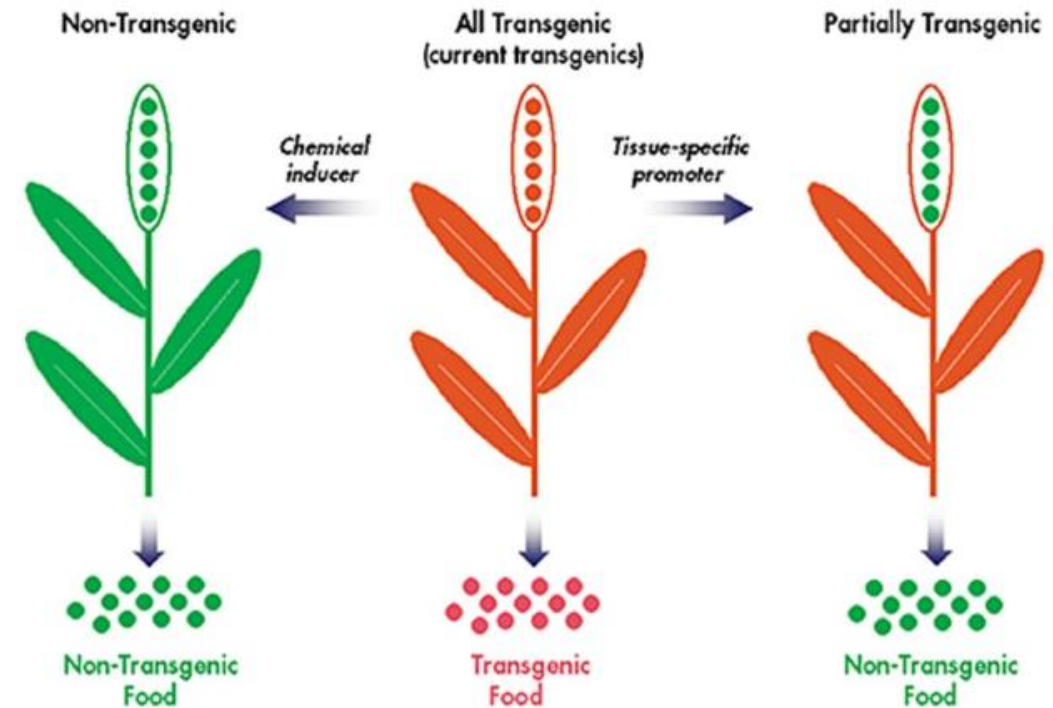
- Transgenic plants are those plants whose genome has been altered.
- They carry foreign gene(s) for creating new properties.
- These new properties may benefit Plant growth, Provide resistance to pests and Diseases or improve the quality of the yield etc.



## BIOTECHNOLOGY

### BENEFITS OF TRANSGENIC CROPS

1. Improvement in Yield.
2. Improvement in nutritional quality (ex – Golden Rice is enriched in Vitamin A) and flavour and taste of yield.
3. Pest and Disease resistance.
4. Herbicide resistance.
5. Resistance to abiotic stresses such as drought, soil salinity, soil acidity, frost etc.
6. Transgenic plants are also used to make products such as biodiesel, biodegradable plastics, medicines etc.

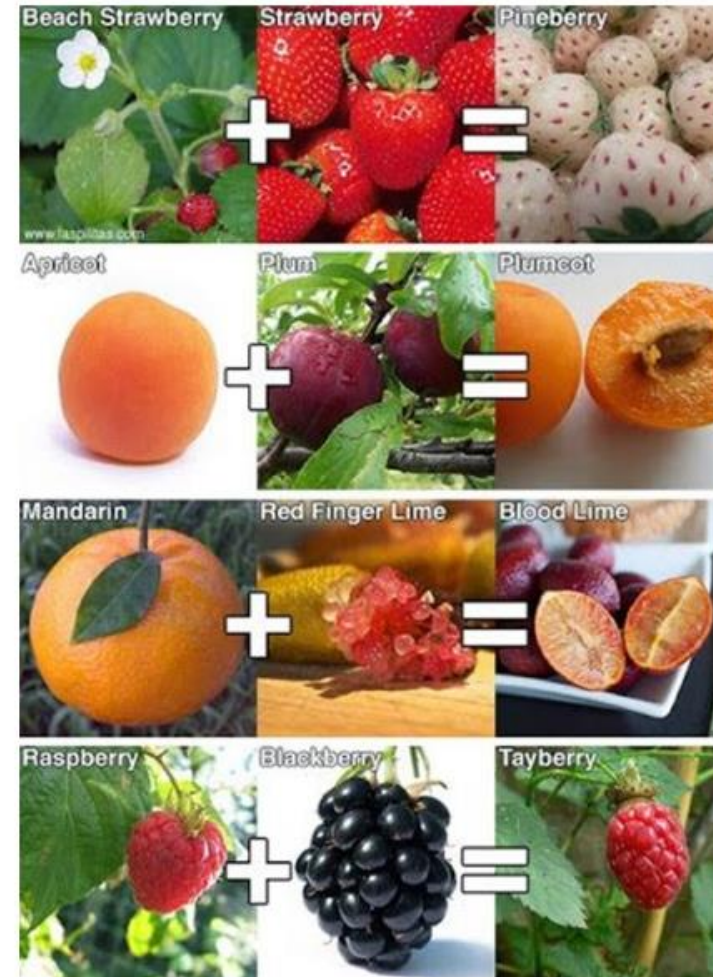




## BIOTECHNOLOGY

### ISSUES WITH TRANSGENIC CROPS

1. Potential adverse impacts on health (several types of allergies may be caused).
2. A concern about transgenic crops causing damage to environment and biodiversity. Example – pollen from transgenic corn kills butterfly larvae.
3. Hybridization of crops with nearby weeds. This could cause weeds to attain resistance to herbicides which could become very problematic.
4. People being uncomfortable with the idea of eating plants which contain DNA that originated from another source such as bacteria or animals.





## GENETICALLY MODIFIED CROPS IN INDIA

The only genetically modified cash crop under commercial cultivation in India is cotton.

## BT COTTON

- It is one of the first GM crop technologies with wide distribution in developing countries. India is one of the largest producers of Bt cotton.
- It is grown over 10.8 million hectares in India, since 2003.
- *Bacillus Thuringiensis* (BT) produces over 200 different toxins which are harmful to several insects. BT toxins are insecticidal but are harmless to other life forms.
- The genes of BT is inserted into cotton, causing the cotton plant to produce natural insecticides.
- The cotton bollworm pest is responsible for huge losses of cotton worldwide. In India losses due to this pest account for 50-60% of the expected yield.





## GENETICALLY MODIFIED CROPS IN INDIA

## BT COTTON

- Traditionally cotton bollworm is controlled using pesticides, but in developing countries like India the expensive pesticides cannot be afforded by the large number of small and marginal farmers.
- So, Bt cotton was developed with the intention of reducing the usage of pesticides, thereby reducing the cost of growing cotton and the impacts on the environment.
- Bt cotton was developed by Monsanto (US).
- Bt cotton is supplied in India by the agri-biotech company Mahyco (in collaboration with Monsanto).





## ISSUES WITH BT COTTON

1. The seeds are more expensive than the local non-GM varieties, hence are not affordable by small and marginal farmers.
2. Monsanto's Bollgard 2 or BG 2 insecticidal technology was supposed to protect the cotton crops against pink bollworm, but the pest has grown resistant to the toxins produced. So, farmers spend more on pesticides to control infestations.
3. International non-profits like Greenpeace have been protesting against Monsanto's monopoly in GM crops. Activists are accusing Monsanto of focussing on its commercial interests while disregarding the issues with GM crops.



# GREENPEACE





## BT BRINJAL

- It was also developed by Mahyco in collaboration with Monsanto.
- Bt Brinjal is resistant to many insects like the Brinjal Shoot and Fruit Borer, which causes a lot of damage to Brinjal crops.
- So, the introduction of Bt Brinjal would lead to increase in the annual production, but there are concerns over the potential health hazards and the problem of terminator seeds which would compel farmers to buy seeds from Monsanto.





## BT BRINJAL

- Bt Brinjal got approval from the Genetic Engineering Appraisal Committee (GEAC) and the government in 2009, making it the first food crop to be approved in India.
- But a lot of concerns were raised and there was stiff opposition.
- The matter was dealt by the supreme court and an expert committee appointed by the SC recommended a 10-year moratorium on all field trials of GMO food crops.
- So, the government imposed a moratorium on field trials of Bt Brinjal in 2010.
- In 2013, Bangladesh gave nod to the commercial cultivation of Bt Brinjal. Experts fear that due to the porosity of Indo-Bangladesh border, Bt Brinjal would leak into India.





## GM MUSTARD

- Dhara Mustard Hybrid II or DMH II is a genetically modified variety of mustard developed by the Delhi University's Centre for Genetic Manipulation of Crops.
- It is a herbicide tolerant crop.
- Scientists say that GM Mustard will give 25-30% more yield than the best varieties.



## CONTROVERSY

- Many scientists claim that GM Mustard will require almost double the quantity of fertilizer and water. There may be potential health impacts too.
- In May 2017, the Genetic Engineering Appraisal Committee (GEAC) declared that GM Mustard passed the tests and recommended to the government to approve it.
- Presently there is no government approval for cultivation of GM Mustard.





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